Math 81: Abstract Algebra

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Problem 2. Let p be an odd prime number, $\zeta = \zeta_p$, and $K = \mathbb{Q}(\zeta)$. We know that $K/\mathbb{Q}$ is a Galois extension with (cyclic) group $G \cong (\mathbb{Z}/p\mathbb{Z})^\times$ and let $\sigma \in G$ be a generator. Let $H \subset G$ be the unique subgroup of index 2. Define

$$\eta_0 = \sum_{\tau \in H} \tau(\zeta), \qquad \eta_1 = \sum_{\tau \in G \setminus H} \tau(\zeta).$$

These are called the periods of ζ with respect to H.

(a) Prove that $\sigma(\eta_0) = \eta_1$ and $\sigma(\eta_1) = \eta_0$ and that

$$\eta_0 = \sum_{a \text{ square}} \zeta^a, \qquad \eta_1 = \sum_{a \text{ nonsquare}} \zeta^a,$$

where the sums are taken over the set of squares and nonsquares, respectively, in $(\mathbb{Z}/p\mathbb{Z})^\times$.

(b) Prove that $\eta_0 + \eta_1 = -1$, and more generally, that

$$\sum_{\tau \in G} \tau(\zeta^a) = -1$$

for any a with $p \nmid a$.

(c) Let

$$g = \sum_{i=0}^{p-1} \zeta^{i^2}$$

be the classical Gauss sum. Prove that

$$g = \sum_{i=0}^{p-2} (-1)^i \sigma^i(\zeta) = \eta_0 - \eta_1.$$

(d) Prove that $\tau(g) = g$ if $\tau \in H$ and $\tau(g) = -g$ if $\tau \in G \setminus H$. Conclude, using the Galois correspondence, that $[\mathbb{Q}(g) : \mathbb{Q}] = 2$. Also conclude that $\overline{g} = g$ if -1 is a square modulo p and that $\overline{g} = -g$ if -1 is not a square modulo p, where the overline is complex conjugation. Hint. For the last part, recall that inversion is the same as complex conjugation for any root of unity.

(e) Prove that $g\overline{g} = p$. Hint. Transform $g\overline{g}$ to the double sum

$$\sum_{k=0}^{p-2}(-1)^k \sum_{j=0}^{p-2} \sigma^j\left(\frac{\sigma^k(\zeta)}{\zeta}\right),$$

then use part (b).

(f) Prove that $g^2 = (-1)^{(p-1)/2}p$.

(g) Finally, conclude that

$$\mathbb{Q}\left(\sqrt{(-1)^{(p-1)/2}p}\right)$$

is the unique quadratic subfield of $\mathbb{Q}(\zeta_p)/\mathbb{Q}$.

Solution.

Note that we know that the set of non-zero squares form an index two subgroup of $\mathbb{F}_p^\times$ (from HW2, Problem 1.5), and $(\mathbb{Z}/p\mathbb{Z})^\times \cong \mathbb{F}_p^\times$. Hence H must be isomorphic to this subgroup.

(a) Let σ be a generator for G. Since H is the subgroup formed by non-zero squares in G, it's elements are of the form σ^{2n}, where $n \in \{1, \cdots, \frac{p-1}{2}\}$, that is they are even powers of σ. So, $\forall \tau \in H$, $\sigma\tau = \sigma^{2n+1} \notin H$. Hence,

$$\sigma(\eta_0) = \sum_{\tau\in H}\sigma(\tau(\zeta)) = \sum_{\tau\in G\setminus H}\tau(\zeta) = \eta_1$$

And similarly, any element τ in $G \setminus H$ is an odd powers of σ. Then $\sigma\tau$ is an even power (odd + 1). Hence, $\sigma\tau \in H$,

$$\sigma(\eta_1) = \sum_{\tau\in G\setminus H}\sigma(\tau(\zeta)) = \sum_{\tau\in H}\tau(\zeta) = \eta_0$$

Moreover since $G \cong (\mathbb{Z}/p\mathbb{Z})^\times$, each $\tau \in G$ corresponds to some $a \in (\mathbb{Z}/p\mathbb{Z})^\times$. Since σ is a generator, let $\tau = \sigma_a$ and we have $\tau(\zeta) = \zeta^a$. And as we already said H has squares, and $G \setminus H$ doesn't. Hence,

$$\eta_0 = \sum_{a \text{ square}} \zeta^a, \qquad \eta_1 = \sum_{a \text{ nonsquare}} \zeta^a,$$

(b) Note that if $p \nmid a$ then ζ^a is not 1. Then,

$$\sum_{\tau\in G}\tau(\zeta^a) = \sum_{b\in(\mathbb{Z}/p\mathbb{Z})^\times}(\zeta^a)^b = \sum_{b\in\{1,\ldots,p-1\}}\zeta^{ab}$$

Since $p \nmid a$, multiplication by a is a bijection on $(\mathbb{Z}/p\mathbb{Z})^\times$. That is as b ranges over $\{1, \dots, p-1\}$ so does $ab \pmod p$. Hence we can write,

$$\sum_{\tau \in G} \tau(\zeta^a) = \sum_{b \in \{1,\dots,p-1\}} \zeta^{ab} = \sum_{c=1}^{p-1} \zeta^c = \zeta \frac{\zeta^{p-1} - 1}{\zeta - 1} = \frac{1 - \zeta}{\zeta - 1} = -1$$

Moreover $\eta_0 + \eta_1 = -1$, when $a = 1$ (because they iterate over H and $G \setminus H$, and hence all of G).

(c) (i) Let $g = \sum_{i=0}^{p-1} \zeta^{i^2}$. Note that as i cycles through $\{1, \dots, p-1\}$, i^2 cycles through every square in that range twice. Hence,

$$\begin{aligned} g &= \sum_{i=0}^{p-1} \zeta^{i^2} \\ &= \zeta^{0^2} + 2 \sum_{a \text{ square}}^{p-1} \zeta^a \\ &= 1 + 2\eta_0 \end{aligned}$$

From part (b) we know $\eta_0 + \eta_1 = -1$, that is $-(\eta_0 + \eta_1) = 1$.

$$g = 1 + 2\eta_0 = -\eta_0 - \eta_1 + 2\eta_0 = \eta_0 - \eta_1$$

(ii) From part (a) we know that η_0 is the sum of elements in G that are even powers of the generator σ applied to ζ, and η_1 is the sum of elements in G that are odd powers of the generator σ applied to ζ. That is,

- Even Powers: σ^0, $\sigma^2, \dots, \sigma^{p-3}$, so $i = 0, 2, \dots, (p-3)/2$.
- Odd Powers: σ^1, $\sigma^3, \dots, \sigma^{p-2}$, so $i = 1, 3, \dots, (p-1)/2$.

So we can write,

$$\begin{aligned} \sum_{i=0}^{(p-3)/2} (-1)^{2i} \sigma^{2i}(\zeta) + \sum_{i=1}^{(p-1)/2} (-1)^{2i-1} \sigma^{2i-1}(\zeta) &= \sum_{i=0}^{(p-3)/2} \sigma^{2i}(\zeta) - \sum_{i=1}^{(p-1)/2} \sigma^{2i-1}(\zeta) \\ &= \sum_{\tau \in H} \tau(\zeta) - \sum_{\tau \in G \setminus H} \tau(\zeta) \\ &= \eta_0 - \eta_1 = g \end{aligned}$$

Hence,

$$g = \sum_{i=0}^{p-2} (-1)^i \sigma^i(\zeta) = \eta_0 - \eta_1.$$

(d) (i) $\tau(g)$, for $\tau \in H$ is

$$\tau(g) = \tau(\eta_0) - \tau(\eta_1)$$

But since τ is an even power of σ, say $\tau = \sigma^{2k}$. We know from part (a) that

$$\sigma(\sigma(\eta_0)) = \sigma(\eta_1) = \eta_0 \qquad \sigma(\sigma(\eta_1)) = \sigma(\eta_0) = \eta_1$$

Thus σ^2 fixes both η_0 and η_1, and therefore any even power σ^{2k} also fixes them. So,

$$\tau(g) = \eta_0 - \eta_1$$

Similarly, $\tau(g)$, for $\tau \in G \setminus H$ is

$$\tau(g) = \tau(\eta_0) - \tau(\eta_1)$$

Then τ is an odd power of σ, say $\tau = \sigma^{2k+1}$ we know from part (a) that

$$\sigma(\eta_0) = \eta_1 \qquad \sigma(\eta_1) = \eta_0$$

Since σ swaps η_0 and η_1 and σ^{2k} fixes them, σ^{2k+1} also swaps them. So,

$$\tau(g) = \eta_1 - \eta_0$$

(ii) That is, g is fixed by an index 2 subgroup of G, i.e. H, and thus by Galois correspondence is a degree two extension of $\mathbb{Q}$. That is,

$$[\mathbb{Q}(g) : \mathbb{Q}] = [G : H] = 2$$

(iii) Note that complex conjugation sends ζ to ζ^{-1}. Thus complex conjugation is the automorphism $\sigma_{-1} \in G$, where

$$\sigma_{-1}(\zeta) = \zeta^{-1}.$$

Hence $\overline{g} = \sigma_{-1}(g)$.
If -1 is a square modulo p, then $a = -1 \in (\mathbb{Z}/p\mathbb{Z})^\times$ is a square, i.e. $\sigma_a \in H$, and by part (i), $\sigma_a(g) = g$.
If -1 is not a square modulo p, then $a = -1 \in (\mathbb{Z}/p\mathbb{Z})^\times$ is not a square, i.e. $\sigma_a \in G \setminus H$, and by part (i), $\sigma_a(g) = -g$.

(e) Note that complex conjugation sends $\sigma_i \to \sigma_{-i}$. So we have,

$$\begin{aligned}
g\overline{g} &= \left(\sum_{i=0}^{p-2} (-1)^i \sigma^i(\zeta)\right)\left(\sum_{j=0}^{p-2} (-1)^{-j} \sigma^{-j}(\zeta)\right) \\
&= \sum_{i=0}^{p-2}\sum_{j=0}^{p-2} (-1)^i(-1)^{-j}\sigma^i(\zeta)\sigma^{-j}(\zeta) \\
&= \sum_{i=0}^{p-2}\sum_{j=0}^{p-2} (-1)^{i-j}\sigma^{i-j}(\zeta)
\end{aligned}$$

Let $k = i - j \pmod{p}$, and note that $\sigma^k(\zeta) = \sigma^j(\sigma^k(\zeta))\sigma^{-j}(\zeta) = \sigma^j\left(\frac{\sigma^k(\zeta)}{\zeta}\right)$. So,

$$\begin{aligned}
g\overline{g} &= \sum_{i=0}^{p-2}\sum_{j=0}^{p-2}(-1)^k\sigma^k(\zeta) \\
&= \sum_{k=0}^{p-2}(-1)^k\sum_{j=0}^{p-2}\sigma^j\left(\frac{\sigma^k(\zeta)}{\zeta}\right)
\end{aligned}$$

If σ corresponds to $a \in (\mathbb{Z}/p\mathbb{Z})^\times$, we have

$$g\overline{g} = \sum_{k=0}^{p-2}(-1)^k\sum_{j=0}^{p-2}\sigma^j(\zeta^{a^k-1})$$

Note that if $k = 0$, then $\zeta^{a^k-1} = 1$, and the inner sum evaluates to $p-1$ otherwise from part (b), the inner sum would evaluate to -1

$$\begin{aligned}
g\overline{g} &= \sum_{k=0}^{p-2}(-1)^k\sum_{j=0}^{p-2}\sigma^j(\zeta^{a^k-1}) \\
&= \sum_{k=0}^{k=0}(-1)^k\sum_{j=0}^{p-2}\sigma^j(\zeta^{a^k-1}) + \sum_{k=1}^{p-2}(-1)^k\sum_{j=0}^{p-2}\sigma^j(\zeta^{a^k-1}) \\
&= (p-1) + \sum_{k=1}^{p-2}(-1)^k(-1) \\
&= (p-1) + (-1)(-1) \\
&= p
\end{aligned}$$

Hence proved.

(f) From part (d) we know that $\overline{g} = g$ if -1 is a square modulo p, and $\overline{g} = -g$ otherwise. Recall from Math 71 that -1 is a square modulo p iff $p - 1 \equiv 0 \pmod{4}$, that is $p-1$ is divisible by 2 twice, and $(p-1)/2$ is even (otherwise it's odd). Hence,

$$g^2 = (-1)^{(p-1)/2}p$$

(g) Note, $\mathbb{Q}(\sqrt{(-1)^{(p-1)/2}p}) = \mathbb{Q}(\sqrt{g^2}) = \mathbb{Q}(g)$. From part (a) we know that $[\mathbb{Q}(g) : \mathbb{Q}] = 2$ and since H is the unique index two subgroup of the galois group G, it fixes this field. Hence, $[\mathbb{Q}(\sqrt{(-1)^{(p-1)/2}p}) : \mathbb{Q}] = 2$.

Problem 3. Fundamental Theorem of Algebra.
An ordered field is a field F together with a subset F^+ of positive elements satisfying: $a, b \in F^+ \Rightarrow a + b \in F^+$ and $ab \in F^+$ and for each $a \in F$ exactly one of $a \in F^+$, $a = 0$, or $-a \in F^+$ is true.

(a) Prove that if F is an ordered field then any nonzero square is positive, that -1 is not positive, and that F has characteristic zero. Also, prove that $F(i) = F[x]/(x^2 + 1)$ is not an ordered field. **Challenge:** Prove that a field F can be ordered if and only if -1 is not a sum of squares.

(b) An ordered field F is called real closed if every positive element has a square root and every polynomial of odd degree over F has a root. Prove that $\mathbb{R}$ and $\mathbb{R} \cap \overline{\mathbb{Q}}$ are real closed. **Hint:** You may need a tiny bit of analysis, but try to keep it to a minimum.

(c) Prove that a real closed field does not have any nontrivial finite extensions of odd degree.

(d) Prove that if F is real closed then the only quadratic extension of F is $F(i)$, and every element of $F(i)$ has a square root.

(e) Prove that a field K is algebraically closed if and only if it does not admit any nontrivial algebraic extensions if and only if it does not admit any nontrivial finite extension.

(f) Prove that if F is a real closed field then $F(i)$ is algebraically closed.

Hint: First, let $L'/F(i)$ be a finite extension and L/F the normal closure of L'/F. Then why is L/F a Galois extension whose group G has even order? Let $H \subset G$ be a Sylow 2-subgroup. Use the Galois correspondence with $H \subset G$ to prove that G is actually a 2-group. Remember the result from abstract algebra that every finite p-group has a subgroup of index p, and use this, with the Galois correspondence, to prove that actually G must be trivial.

(g) Deduce that $\mathbb{C}$ and $\overline{\mathbb{Q}}$ are algebraically closed.

Solution.

(a) Let F be an ordered field. Then,

(i) Any non-zero square is positive.
Let a be a non-zero element in F. If $a \in F^+$, then $a^2 = a \cdot a \in F^+$.
If $-a \in F^+$, then $(-a)^2 = (-a) \cdot (-a) \in F^+$. We want to show that $(-a)^2 = a^2$.
Since, $-a = -1 \cdot a$, this is equivalent to showing that $-1^2 = 1$ in F.

Now note that,

$$(-1)(-1)+(-1)1=(-1)(-1+1)=0$$

But that also implies,

$$(-1)(-1)=-(-1)=1$$

Hence, proved.

(ii) -1 is not positive.

We will show this by contradiction. Let $p \neq -1$, $p \in F^+$. For the sake of contradiction, assume $(-1) \in F^+$. Then since for any $a, b \in F^+, ab \in F^+$, $p(-1) = -p \in F^+$. However, by definition of an ordered field, both $-p$ and p can't be in F^+. Hence a contradiction, and -1 cannot be positive.

(iii) F has characteristic 0.

The characteristic of a field F is defined as the smallest positive integer n such that $1+1+\cdots+1=0$ (n times) in F. From part (ii), we know that $-(-1)=1 \in F^+$, $0 \notin F^+$, and we that for any $a, b \in F^+$, $a+b \in F^+$. So, $1+\cdots+1$ (n times), for any n cannot equal 0. Hence, such an integer doesn't exist and F must be characteristic 0.

(iv) $F(i) = F[x]/(x^2+1)$ is not an ordered field.

We will show this by contradiction. Assume $F(i)$ is an ordered field. Then for any $a \in F(i)$, exactly one of $a \in F^+, -a \in F^-, a = 0$ Then in $F(i)$, $i^2+1=0 \implies i^2=-1$, but from part (i) we know that non-zero squares are positive, and from part (ii) we know that -1 is not positive. So, i must be 0, but we also know $(0\cdot 0 = 0(0+0) = 0\cdot 0 + 0 \cdot 0 \implies 0 \cdot 0 = 0\cdot 0 - 0 \cdot 0 = 0)$ that $0^2 = 0$. Hence, $i \notin F^+, -i \notin F^+, i \neq 0$, which is a contradiction, and $F(i)$ cannot be an ordered field.

(v) F can be ordered if and only if -1 is not a sum of squares.

($\implies$) Assume F is ordered.

Let $a, b \in F$. Then, from part (i) we know that $a^2, b^2 \in F^+$, that is $a^2+b^2 \in F^+$. But since $-1 \notin F^+$, it cannot be the sum of squares.

($\impliedby$) Assume -1 is not a sum of squares in F.

Definition: A field is called a real field if -1 is not a sum of two sqaures in the field.

Definition: A field is real closed if it is real and has no proper, real extensions.

Claim: Every real field has a real closure.

Proof: Let F be a real field, K_i/F, for $i \in$ the indexing set I, be algebraic extensions of F. Then, we define the set $K = \{K_i : K_i/F$ real algrbriac extensions of $F\}$. Then we want to find M such that $\forall K_i \in K, K_i \subseteq M$. Let L be such that,

$$L = \bigcup_{K_i \in K} K_i$$

By Zorn's Lemma, if at least one K_i is non-empty, and every constituent chain $C \subseteq L$ has an upperbound this limit exists.

i. L is not empty since F/F is a real algebraic extension of F.

ii. Every constituent chain $C \subseteq L$ has an upperbound.
Let $K' = \bigcup_{K_i \in C} K_i \in C$ be this upper bound. We show that K' is a real field.

- Closed under Inverses:
If $a \in K'$ then there exists some $K_i \in C$ such that $a \in K_i$. Since K_i is a field, both the additive $-a$ and multiplicative inverse a^{-1} of a is in K_i and hence in K'.
- Closed under Addition:
$\forall a, b \in K'$, then there exist K_i, $K_j \in C$ such that $a \in K_i$, $b \in K_j$. Since C is totally ordered, either $K_i \subseteq K_j$ or $K_j \subseteq K_i$. Without loss of generality, asssume $K_i \subseteq K_j$, then $a \in K_j$, and since K_j is a field, it is closed under addition, and there exist a $c \in K_j \subseteq K'$ such that $a + b = c$. Hence K' is closed under addition.
- Closed under Multiplication:
Similarly, $\forall a, b \in K'$, then there exist K_i, $K_j \in C$ such that $a \in K_i$, $b \in K_j$. Since C is totally ordered, either $K_i \subseteq K_j$ or $K_j \subseteq K_i$. Without loss of generality, asssume $K_i \subseteq K_j$, then $a \in K_j$, and since K_j is a field, it is closed under multiplication, and there exist a $c \in K_j \subseteq K'$ such that $a \cdot b = c$. Hence K' is closed under multiplication.
- Distributivity, Commutativity, and Associativity also work similarly.
- -1 is not a sum of two squares:
For the sake of contradiction assume it is. Then there exist some $a, b \in K'$ such that $a^2 + b^2 = -1$. Thus, there exist K_i, $K_j \in C$ such that $a \in K_i$, $b \in K_j$. Since C is totally ordered, either $K_i \subseteq K_j$ or $K_j \subseteq K_i$. Without loss of generality, asssume $K_i \subseteq K_j$, then $a \in K_j$. Then $a^2 + b^2$ should sum to -1, but K_j is real, a contradiction. Hence, K' must be real.

Hence, K' is a real field, and L exists. Thus, every field has a real closure.

Claim: A real closed field is ordered.

Proof: Let F be a real closed field. Then F has no real, proper, algebraic extensions. n particular, for every $a \in F$, exactly one of the polynomials

$$f(x) = x^2 - a, \qquad g(x) = x^2 + a$$

has a root in F.

Since F is real, -1 cannot be written as a sum of squares. Hence $x^2 + 1$ has no root in F, so $\sqrt{-1} \notin F$. It follows that for every $a \neq 0$, either a is a square in F or $-a$ is a square in F. Now, we can establish an ordering for F.

Let F^+ be the set,

$$F^+ = \{a : \sqrt{a} \in F\}$$

i. F^+ is closed under addition.
$\forall a, b \in F^+$, then both a, b are squares and $a + b$ satisfies the polynomial $f(x) = x^2 - (a + b)$. Since F has no real, proper, algebraic extensions $\sqrt{a + b}$ must be in F and hence $a + b \in F^+$.

ii. F^+ is closed under multiplication.
$\forall a, b \in F^+$ if $a = x^2$, $b = y^2$ then $a \cdot b = x^2 \cdot y^2 = (xy)^2$. Hence, $ab \in F^+$.

iii. For all $a \in F$ either $a = 0$ or $a \in F^+$ or $-a \in F^+$.
From what we said earlier, if $a \in F^+$ then $-a \notin F^+$. Otherwise both a and $-a$ would be squares, so

$$-1 = \frac{-a}{a}$$

would be a square in F, implying $\sqrt{-1} \in F$, contradicting that F is real.
And, if $a \notin F^+$ then a is not a square. By the property established earlier, this implies $-a$ is a square, hence $-a \in F^+$.

Hence F^+ forms the positive subset of F and defines an ordering on F.

Claim: All real fields are ordered.

Proof: Let F be a real field, and $\overline{F}$ be it's real closure. From the claim above we know that $\overline{F}$ is ordered, i.e., there exists a positive subset $\overline{F}^+ \subset \overline{F}$. Then, consider the subset $F^+ = F \cap \overline{F}^+$. We claim that this is the positive subset that defines an order on F.

i. F^+ is closed under addition.
If $a, b \in F^+$ then $a, b \in \overline{F}^+$. And since $\overline{F}^+$ is closed under addition $a + b \in \overline{F}^+$. But because F is a field, $a + b \in F$. Hence $a + b \in F \cap \overline{F}^+$. Since a, b were arbitrary this works for all $a, b \in F^+$.

ii. F^+ is closed under multiplication.
Similarly, if $a, b \in F^+$ then $a, b \in \overline{F}^+$. And since $\overline{F}^+$ is closed under multiplication $ab \in \overline{F}^+$. But because F is a field, $ab \in F$. Hence $ab \in F \cap \overline{F}^+$. Since a, b were arbitrary this works for all $a, b \in F^+$.

iii. For all $a \in F$ either $a = 0$ or $a \in F^+$ or $-a \in F^+$.
Since $\overline{F}$ is ordered, for any $a \in F^+$, $a \neq 0$, $\overline{F}^+$ only contains one of a and $-a$. Hence, so does F^+.

Thus, F^+ forms the positive subset of F and defines an ordering on F. Hence, all real fields are ordered.

Thus, a field F can be ordered if and only if -1 is not a sum of squares.

(b) (i) $\mathbb{R}$ is real-closed.

Let $a \in \mathbb{R}^+$. Finding the squareroot of a is equivalent to solving the quadratic polynomial $f(x) = x^2 - a$ over $\mathbb{R}$. Now, note that at $x = 0$, $f(x) = -a < 0$ and when $x > a$ then $f(x) > a > 0$. Then, by the intemediate value theorem there must exist some x such that $f(x) = 0$. Hence, every positive real number has a square root.

Let $f(x) \in \mathbb{R}$ be a monic polynomial of degree n, where n is odd. Then as $x \to \infty$, $f(x) \to \infty$ and as $x \to -\infty$, $f(x) \to -\infty$. So there exist some a, b such that $f(a) < 0 < f(b)$. Then by the intermediate value theorem there must exist some c such that $f(x) = 0$. Hence every odd degree polynomial has at least one real root.

Also note that over $\mathbb{C}$ non-real roots come in pairs. Since $f(x) \in \mathbb{R}$, complex conjugation fixes all coefficients of f. So if α is a root of $f(x)$, the complex conjugate of $f(\alpha)$ is $\overline{f(\alpha)} = f(\overline{\alpha}) = 0$ Hence, $\overline{\alpha}$ is also a root. That is, complex roots come in pairs.

(ii) $\mathbb{R} \cap \overline{\mathbb{Q}}$ is real-closed.

Recall that $\overline{\mathbb{Q}}$ is the algebriac closure of $\mathbb{Q}$. So, $\overline{\mathbb{Q}} \cap \mathbb{R}$ is the real part of the algrbriac closure of the rationals. For convenience, let $K = \overline{\mathbb{Q}} \cap \mathbb{R}$.

Let $a \in K^+$. From part (i) we know that all positive real numbers have square roots, that is $\sqrt{a} \in \mathbb{R}$ So, it suffices to show that the squareroots is contained in $\overline{\mathbb{Q}}$. Since $\overline{\mathbb{Q}}$ is the algrbriac closure of $\mathbb{Q}$ and our subfield contains all real algebriac roots to polynomials over $\mathbb{Q}$, a must satisfy some polynomial $f(x)$ over $\mathbb{Q}$. Then note that $\sqrt{a}$ satisfies $f(x^2)$. Hence $\sqrt{a}$ is algebriac and must be in the field.

Let $f(x) \in K[x]$ be a polynomial of odd degree n. Note that if we consider $f(x) \in \overline{\mathbb{Q}}$, then all roots of $f(x)$ are in $\overline{\mathbb{Q}}$. Since K is a restriction of $\overline{\mathbb{Q}}$ to the reals, and complex roots come in pairs, K at least has one root that satisfies $f(x)$. Hence every odd degree polynomial has a root.

(c) Let F be a real-closed field, and K/F a finite extension. We know that finite extensions are generated by a finite number of algebraic elements over the base field. Let K/F be generated by m elements. Then there exists a tower,

$$\begin{array}{c} K = F(\alpha_1, \dots, \alpha_{m-1})(\alpha_m) \\ \big\uparrow \\ K = F(\alpha_1, \dots, \alpha_{m-2})(\alpha_{m-1}) \\ \big\uparrow \\ \vdots \\ \big\uparrow \\ F(\alpha_1) \\ \big\uparrow \\ F \end{array}$$

Let $f_i(x) = m_{\alpha_i/F(\alpha_1,\dots,\alpha_{i-1})}(x)$ be the minimal polynomial of degree n_i of such an algebraic element. Note that n_i is either 1 (trivial extension) or even because from part (ii) we know that polynomials of odd degree have a root over the base field, that is if n_i is odd then $f_i(x)$ is not irreducible and hence not the minimal polynomial. Also since the degree of the minimal polynomial is the degree of the extension, and each $f_i(x)$ has even degree, and by tower law $[K : F] = \prod_{i=1}^{m} n_i$ which is a product of even numbers and hence even. Hence, F has no nontrivial finite extensions of odd degree.

(d) Let F be real-closed. Let K/F be a quadratic extension. We know that K is generated by an element that satisfies a quadratic minimal polynomial $f(x) \in F[x]$. But by definition, all positive elements have a square root in F. So, $f(x)$ must be satisfied by the squareroot of a negative element in F. But since F is a field, we know that -1 exists, and any negative element $a \in F$ (that is $-a \in F^+$) can be written as $a = -1 \cdot -a$. Since $\sqrt{-a} \in F$, adjoining the squareroot of a is equivalent to adjoining $\sqrt{-1}$. Hence, every quadratic extension of F is $F(i)$, where $i = \sqrt{-1}$.

Let $(a+bi)$ be an arbitrary element in $F(i)$, where $a, b \in F$. If $b = 0$ then $\sqrt{(a+bi)} = \sqrt{a} \in F$. If $b \neq 0$, we want to determine if there exist $c, d \in F(i)$ such that $(c + di)^2 = (a + bi)$.

On expanding we get,

$$c^2 - d^2 = a, \qquad 2cd = b$$

Now, note that $(c^2 + d^2)^2 = (c^2 - d^2)^2 + (2cd)^2 = a^2 + b^2$. And, $c^2 + d^2 = \sqrt{a^2 + b^2} \in F$ since F is real-closed. Combining this with $c^2 - d^2 = a$ gives us,

$$c^2 = \frac{\sqrt{a^2 + b^2} + a}{2}, \qquad d^2 = \frac{\sqrt{a^2 + b^2} - a}{2}$$

which are both positive since $\sqrt{a^2 + b^2} \geq \sqrt{a^2} \geq |a| > 0$. Hence, their square-roots exist in F, and we found are desired squareroot.

(e) Let $A : K$ is algebraically closed, $B : K$ doesn't admit any non-trivial algebraic extensions, $C : K$ does not admit any non-trivial finite extension.

$A \implies B$

Assume K is algebraically closed. Then any polynomial with coefficients in K has roots in K. That is, there are no non-trivial algebraic extensions.

$B \implies C$

Assume K doesn't admit any non-trivial algebraic extensions. We know that all finite extensions are algebraic. Then, since K doesn't admit any non-trivial algebraic extensions, it doesn't admit any non-trivial finite extensions.

$C \implies A$

Assume K doesn't admit any non-trivial finite extensions. For the sake of contradiction, assume K is not algebraically closed. Then there exists at least one polynomial $f(x) \in K[x]$, such that $\alpha \notin K$ is a root. Then $K(\alpha)/K$ is a simple finite extension. But K doesn't admit any non-trivial finite extensions, so $K(\alpha)$ must be a trivial extension, i.e., $\alpha \in K$. A contradiction! Hence, K is algebraically closed.

(f) Let F be a real closed field and $F(i) = F(\sqrt{-1})$. Now let $L'/F(i)$ be a finite extension, and L/F be the normal closure of L'/F.

$$\begin{tikzcd}
L \\
L' \arrow[u] \\
F(i) \arrow[u, "n"] \\
F \arrow[u, "2"] \arrow[uu, bend right, "p"'] \arrow[uuu, bend right=60, "m"']
\end{tikzcd}$$

Since F is characteristic 0, all of it's extensions must be characteristic 0, and hence all polynomials over L are separable. And since normal closures are algrbriac, we know that L/F is a normal, separable, and algebraic extension and hence is Galois. Let $G = Gal(L/F)$. Note that from part (c) we know that $[L : F] = m$ is even and hence $|G|$ is even. Then we claim that G is a 2-group.

Proof: Assume for the sake of contradiction that G is not a 2-group. Let $|G| = m = 2^x \cdot y_1 \cdot \ldots \cdot y_n$, where $y_1, \cdots, y_n$ are odd primes. Then by Sylow's Theorems, we know that there exist subgroups of order $y_1, \cdots, y_n$ in G. Then by Galois correspondence, there exist subextensions of F of degrees $y_1, \cdots, y_n$, but since F has no nontrivial finite extensions of odd degree (part (c)) this is a contradiction. Hence, G must be a 2-group.

$\mathsf{Gal}(\mathsf{L}/\mathsf{F}(\mathfrak{i}))$ is a subgroup of the 2-group G, hence is itself a 2-group. If it were nontrivial, every nontrivial $\mathfrak{p}$-group has a subgroup of index $\mathfrak{p}$, so$\mathsf{Gal}(\mathsf{L}/\mathsf{F}(\mathfrak{i}))$ would have a subgroup of index 2. By Galois correspondence this yields a degree-2 extension of $\mathsf{F}(\mathfrak{i})$, contradicting part (d). Hence $\mathsf{Gal}(\mathsf{L}/\mathsf{F}(\mathfrak{i}))$ is trivial, and $\mathsf{F}(\mathfrak{i})$ is algebraically closed.

(g) $\mathbb{R}$ and $\overline{\mathbb{Q}} \cap \mathbb{R}$ are real closed fields, and from part (f) we know that $\mathbb{R}(\mathfrak{i}) = \mathbb{C}$ and $\overline{\mathbb{Q}} \cap \mathbb{R}(\mathfrak{i}) = \overline{\mathbb{Q}}$ are algebraically closed. Hence, $\mathbb{C}$ and $\overline{\mathbb{Q}}$ are algebraically closed.